Below is a comprehensive, professional Model Risk Management (MRM) documentation template you can use for any model—AI/ML, statistical, financial, or operational. It aligns with common regulatory expectations (e.g., OCC 2011-12 / SR 11-7–style frameworks) and modern AI governance standards.

Model Risk Management Documentation Template

1. Model Overview

1.1 Model Name

0422026_Sid_NVDA_Ndocs_RAG

1.2 Model ID / Version

v0

1.3 Date Created / Last Updated

4/22/206

1.4 Model Owner

Siddhant Agarwal

1.5 Model Developer(s)

Siddhant Agarwal

1.6 Model Reviewer(s)

Katherine Stout

Jainesh Lad

1.7 Model Users / Stakeholders

Internal company, work employees.
Company search leadership team, CVP, Exec sponsor

1.8 Purpose & Business Use Case

There are a lot of enterprise content, data. As we explore AI solutions, it is crucial to have a robust RAG system that can retrieve and share Enterprise specific knowledge with the users as grounding. The goal of this RAG model is to ground our internal company chats in our own knowledge.

1.9 Model Category / Type

Supervise ML Gradient Boosting / Tree based model

1.10 Criticality / Risk Tiering

Medium - The risk of showing irrelevant company data, showing outdated information, impeding productivity of employees.

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2. Model Scope & Boundaries

2.1 In-Scope Components

Describe components included in the model boundary:

- Data pipeline
  - Having a corpus of company data - NVDocs
  - Creating Question, Answer, Chunk pairs. This is our training/label. The model trains on for a given query, what is best chunk/doc. Ideal chunk/docs are target = 1, among irrelevant chunks/doc = 0.
  - Initially, we started with irrelevant chunks being picked randomly (outside of targets), however we moved towards selecting irrelevant chunks from top k BM25 scores to help add “good” negative labels.

Example:

```
Question: How does the utilization of GPU acceleration in GATSPI
contribute to its significant speedup over commercial simulators,...
  Chunk ID: 0
  Chunk text: In this paper, we present GATSPI, a novel GPU accelerated
logic gate simulator that enables ultra-fast power estimation ...
  Relevance: 1
```

- Feature engineering
  - I did one-hot encoding for some of the categorical features (e.g. reasoning type, question type)
  - Created BM25 score, cosine similarity, token_overlap_ratio, etc
  - To help normalize some of the scores, I also turned them into cardinal values (e.g. BM25: 4.7 → BM25 Ranking: 1)
- Algorithm(s)

    - Based on Evaluation, the model selected is a LightGBM. However, we also evaluated other candidates. XGBoost, Random Forest, BM25 Baseline, Decision Tree, TF-IDF Cosine
    - We are also computing TF-IDF, BM25 (which is similar to TF-IDF however it compensates for longer docs naturally having more words) for all the documents in the corpus/index
- Post-processing
    - No post processing - ranked results are served to the user. Other than potential UI/UX treatment
- Monitoring dashboards
    - We could monitor the production NDCG@10 and MRR (Mean Rec Rank). Recall@10, Precision@10, or F1 score

2.2 Out-of-Scope Components

N/A

2.3 Dependencies & Interfaces

N/A

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3. Data Documentation

3.1 Data Sources

NVDocs is our data source. In an application setting, the tenant/organization admin or IT department would be responsible for maintaining the corpus. They would manage permissions, ACLs, etc.

3.2 Data Quality Assessment

Given this was synthetic data from NVDA, there were no missing values. However, when setting up our Question/Document/Chunk pairs, we did find some questions only had the expected document and not chunk. We skipped these in training as it is assumed it is noise or unexpected.

3.3 Data Preprocessing

Since we are largely dealing with text data, we did not do a lot of pre-processing. This was to preserve the semantic meaning and avoid removing terms that could be helpful.

During training, as mentioned above we have a rule that samples must contain the question/document/chunk ids.

There were categorical features, such as reasoning type & question type. These were one-hot encoded given the limited possible values.

3.4 Training / Testing / Validation Samples

The dataset is very large, especially after chunking. I originally tried to do a 80 train, 10 Val, 10 Test split. However, this ended up with ~200K chunks X ~85k Question Answer pairs in the train set. Hence, we sampled the train further down to 10k samples. Similar to the size of val, test in the end.

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4. Methodology & Model Design

4.1 Conceptual Design

We are using a tree based model called LightGBM. It is building off the gradient boosting trees (e.g. XGBoost) but with faster training times and less memory usage.

4.2 Algorithm Selection Rationale

We chose to turn the problem into a decision-tree (0 or 1 golden label) for aiming to have the best accuracy and largest pool of positive labels. During our evaluation, we experimented with XGBoost, LightGBM, Decision Tree, Random Forest, and then raw TF-IDF & BM25

4.3 Model Architecture

Include diagrams if applicable.

4.4 Feature Engineering

Feature	Definition	Rationale
`bm25_score`	BM25 relevance score (k1=1.5, b=0.75)	Term frequency to doc frequency compensating for longer documents naturally having more words
`cosine_similarity`	TF-IDF cosine similarity	Term frequency to Doc frequency
`token_overlap_ratio`	fraction of query tokens found in chunk	Simple lexical match measure
`token_overlap_count`	raw count of shared tokens between query and chunk	Absolute overlap (length-independent)
`question_length`	Word count of question	Longer queries may need different ranking behavior
`chunk_word_count`	Word count of chunk (from dataset metadata)	Controls for chunk size effects on scoring

`chunk_sentence_count`	Sentence count of chunk (from dataset metadata)	Structural complexity of chunk
`question_complexity`	Complexity label from QA metadata	Proxy for retrieval difficulty
`hop_count`	Number of reasoning hops from QA metadata	Multi-hop questions need multiple chunks
`bm25_rank`	Per-question rank by BM25 score	Relative position of BM25. Normalize the raw rank score.
`tfidf_rank`	Per-question rank by TF-IDF cosine	Relative position from TF-IDF. Normalize the raw rank score.
`bm25_reciprocal_rank`	1 / bm25_rank	Emphasizes top-ranked candidates nonlinearly
`rank_diff`	bm25_rank − tfidf_rank	Disagreement between BM25 - TF-IDF

We dropped all the reasoning, query type features.

```
Feature Importance:

              feature  importance
   cosine_similarity    0.300484
          bm25_score    0.262873
          tfidf_rank    0.127438
bm25_reciprocal_rank    0.120566
           bm25_rank    0.094456
 token_overlap_ratio    0.045561
 token_overlap_count    0.016807
           rank_diff    0.010576
           hop_count    0.006085
    chunk_word_count    0.005242
     question_length    0.003945
chunk_sentence_count    0.001156
     qtype_multi_hop    0.000933
 question_complexity    0.000681
    qtype_contextual    0.000483
    qtype_structural    0.000433
    rtype_relational    0.000363
        rtype_causal    0.000360
    rtype_procedural    0.000354
   rtype_inferential    0.000352
      rtype_temporal    0.000312
       rtype_factual    0.000306
        rtype_visual    0.000234
```

4.5 Model Assumptions

N/A

4.6 Limitations

- Due to the limited amount of compute, memory I did not try embedding based or more powerful models
- We are taking a sample of training data
- Currently only evaluated on English

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5. Model Performance

5.1 Training Performance Metrics

To measure performance, we are using NDCG@10 as our guiding metric, along with Recall@10, Mean Reciprocal Rank, Mean Average Precision

5.2 Validation Performance

The best model was LightGBM, however overall the models performed similarly. The TF-IDF baseline performed the worst by far, however other models performed similarly.

Model	NDCG@10	Recall@10	MRR	MAP
BM25 Baseline	**0.5227**	**0.5367**	**0.5694**	**0.4747**
TF-IDF Baseline	**0.3487**	**0.3897**	**0.3821**	**0.2998**
Decision Tree	**0.5135**	**0.5216**	**0.5552**	**0.4655**
Random Forest	**0.5335**	**0.5429**	**0.5719**	**0.4858**
XGBoost Ranker	**0.5355**	**0.5440**	**0.5738**	**0.4878**
LightGBM Ranker	**0.5372**	**0.5464**	**0.5744**	**0.4886**

5.3 Backtesting (if relevant)

N/A

5.4 Sensitivity Analysis

N/A - data is synthetic

5.5 Benchmarking

Comparison	NDCG@10	Delta vs BM25
BM25 (champion baseline)	0.5227	—
TF-IDF	0.3487	−33.3%
Best tree model (LightGBM)	0.5372	+1.5%

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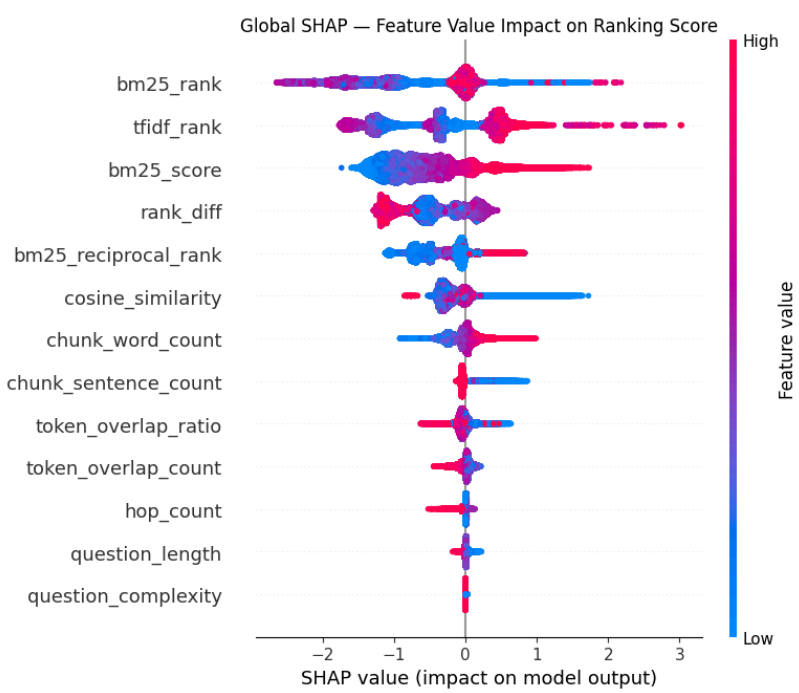
6. Explainability & Interpretability

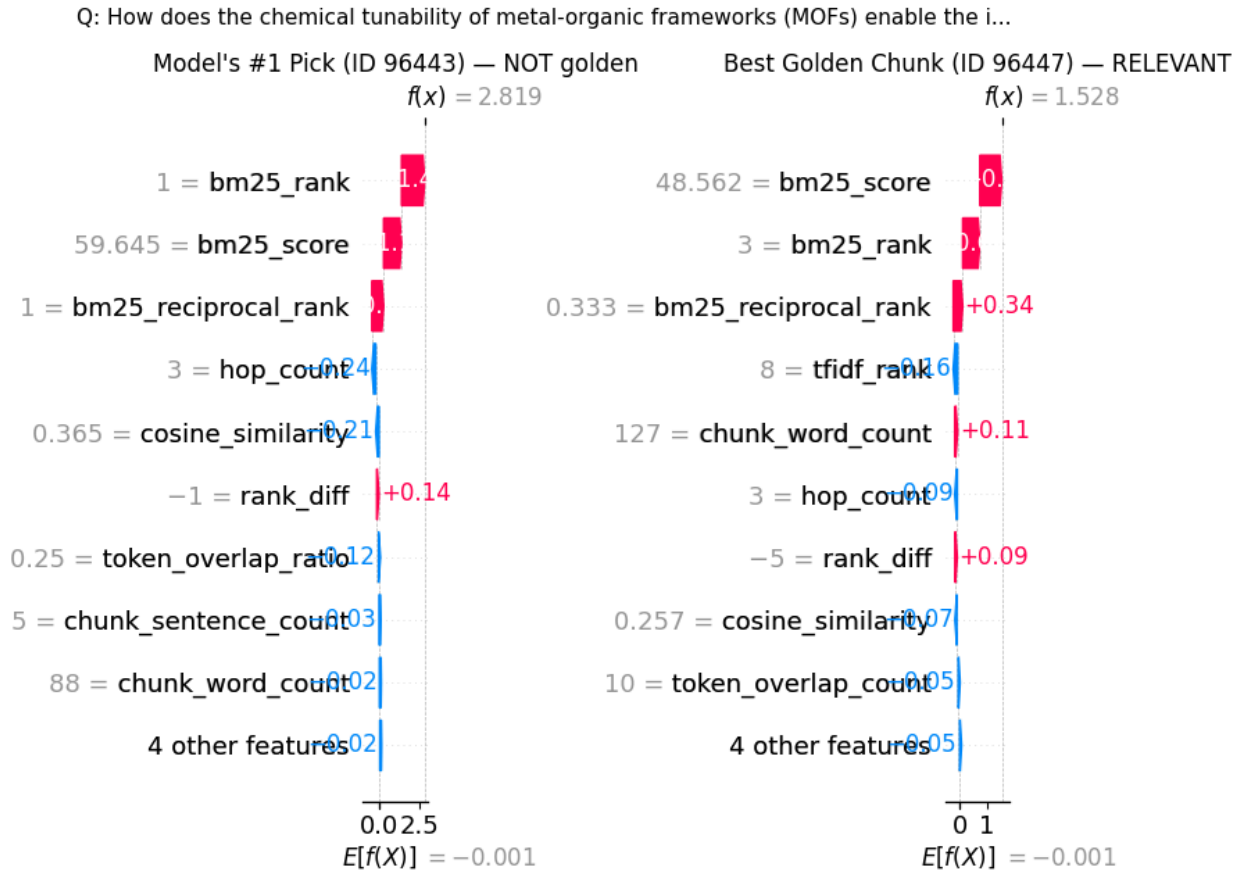
6.1 Global Explainability

I used SHAPley to show the highest impact features. The BM25_rank, TF_IDF rank based features had the highest value. These features are normalized versions of the raw scores.



6.2 Local Explainability

Trying an example question below, we see the BM25 rank and score are the highest value features on both the Models #1 pick (which is not the golden label) and the expected golden label. Judging by the Chunk ID, we are showing the right document but likely not the exact passage.



6.3 Business Interpretability Assessment

Yes, the model should be understandable and explainable from a business POV. We have a very limited # of features, 14. With SHAP values for a given query and overall model we can see what features have the biggest impact on a document ranking at the position it does.

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7. Fairness, Bias, and Ethical Assessment

7.1 Protected Attributes Reviewed

N/A - We don’t have an PII or anything.

7.2 Bias Metrics Evaluated

N/A

7.3 Fairness Mitigation Approaches

N/A

7.4 Ethical Considerations

Potential harms - If the results are being leveraged by a chatbot, and no results are returned. The LLM may hallucinate. We should prompt engineer + add safeguards in model having some deterministic action when no results are returned by RAG.

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8. Model Controls & Governance

8.1 Access & Permission Controls

The RAG Data Science Team - Me

8.2 Change Management

We will use Github for tracking all changes, history, versions. etc.

8.3 Model Inventory Registration

Each model will have a different version with the date it was created.

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9. Model Monitoring Strategy

9.1 Performance Monitoring

- To track model performance, we can rely on NDCG@10, Precision@10, Recall@10
- We should also implement some user thumbs up/thumbs down for measuring user satisfaction scores

9.2 Data Drift & Stability Monitoring

- We can continue to track over time # of items in the corpus, # of chunks per document to learn about how the corpus and underlying data is changing
- We can monitor NDCG@10, Customer Satisfaction scores for stability.

9.3 Alerts & Escalation Procedures

9.4 Periodic Review Requirements

We will review the success metrics quarterly and corpus data for retraining rankers cadence.

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10. Validation Summary - NOT NECESSARY FOR PROJECT

(Completed by independent validator or internal MRM function.)

10.1 Validation Scope

10.2 Validation Findings

- Strengths
- Weaknesses
- Remediation items

10.3 Model Risk Rating

10.4 Validation Conclusion / Recommendation

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11. Compliance & Regulatory Considerations - NOT NECESSARY PROJECT

- Relevant regulatory frameworks
- Compliance controls
- Data protection considerations (GDPR, DPDP, HIPAA, etc.)

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12. Documentation Appendix

- Diagrams
- Technical specifications
- Code snippets (non-sensitive)
- Glossary
- Acronyms
- Testing logs
- Decision logs